

CUSTOMER MILESTONE BRIEF

MS #11 - Hardened Admitted Execution

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.11.0, 0.11.1, 0.11.2
Date	2026-08-11 to 2026-08-11
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #11 - Hardened Admitted Execution converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Moves factory execution from policy simulation to signed, identity-bound, constrained container runs.

Why this matters to a customer

Moves factory execution from policy simulation to signed, identity-bound, constrained container runs.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added signed, expiring work-order envelopes with trusted issuer keys, exact work-order and execution-policy identities, minimum key strength, maximum TTL, and append-only nonce replay prevention.
- Added short-lived, work-order-bound, audience- and action-scoped credentials for planner, builder, provider, and verifier roles; wrong role, action, work order, policy, signature, audience, time window, or issuer fails closed.
- Added an allowlisted one-use protected-value broker whose receipts never contain values and whose in-memory leases clear their contents after consumption.
- Added a real Docker/Podman backend with a digest-pinned image, network disabled, read-only root and workspace mounts, numeric non-root identity, all capabilities dropped, no-new-privileges, process/memory/CPU/tmpfs limits, environment filtering, command allowlisting, no shell, bounded output, and timeouts.
- Added a deterministic offline conformance receipt and separate live enforcement probe. Simulation is explicitly non-production-ready and cannot satisfy hardened-execution promotion policy.
- Integrated admission, agent-identity and gate-execution evidence into factory ledgers and run receipts, plus execution posture in the Factory control room.

- Added a non-overridable hardened-execution audit decision, Evidence Control Tower projection, and release-dossier gate alongside runtime and mainframe evidence.
- Added cross-platform execution launchers, three JSON Schema contracts, policy weakening, tampering, expiry, replay, identity scope, protected-value leakage, OCI invocation, orchestration, API and UI tests, and full-suite verification.
- Split execution assurance into deterministic policy simulation, live container-runtime probe, and signed admitted OCI factory-run evidence; a probe can no longer satisfy the factory gate.
- Bound work-order admission, exact policy and work-order identities, issued agent identities, attested role actions, enforced acceptance gates, and non-persistent protected values into the factory execution-security receipt.
- Added strict evidence normalization that recomputes readiness, rejects unknown receipt types, invalid hashes, partial bindings, failed gates, missing actions, and producer-asserted readiness.
- Added a one-command macOS/Linux and Windows admitted-run workflow that signs, executes through Docker/Podman, validates the composite receipt, and builds a live audit snapshot and dossier.
- Updated audit ingestion, non-overridable policy, release dossier, Control Tower, and Factory UI to distinguish evidence class and clear only hardened execution while z/OS equivalence remains blocked.
- Added a normalized execution-evidence schema and adversarial tests proving that a successful live probe cannot impersonate a signed factory run.
- Corrected OCI environment translation so the host workspace path becomes /workspace inside Docker/Podman, matching the read-only bind mount used by private acceptance gates.
- Added a regression assertion that rejects leaking the host workspace path into the container environment while preserving the fixed /workspace/src Python module path.
- Retained the v0.11.1 signed-admission and composite-evidence contract unchanged.

How it differs from earlier work

MS #11 builds on MS #10 (Audit Ledger and Evidence Control Tower). The earlier milestone established its own bounded capability; this milestone adds hardened admitted execution while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use hardened admitted execution as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- Added a deterministic offline conformance receipt and separate live enforcement probe. Simulation is explicitly non-production-ready and cannot satisfy hardened-execution promotion policy.
- Updated audit ingestion, non-overridable policy, release dossier, Control Tower, and Factory UI to distinguish evidence class and clear only hardened execution while z/OS equivalence remains blocked.

Source of record

- CHANGELOG.md release section(s): 0.11.0, 0.11.1, 0.11.2
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity